NIBS-SES

Non-Invasive Brain Stimulation Side Effect Scale

User Manual & Administration Guide

The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES) is a free, quick, multi-language screening of adverse effects in transcranial magnetic, electrical and ultrasound stimulation

Manual version **1.2** · Instrument version **1.1**

Official site: www.nibs-ses.com · Repository: osf.io/xz3nb

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1. About the NIBS-SES

The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES) is a free, quick, multi-language self-report questionnaire for screening the adverse effects of non-invasive brain stimulation. Its distinguishing feature is that a single instrument covers all three major stimulation modalities, comprising transcranial magnetic stimulation (TMS, including rTMS, iTBS, and cTBS), transcranial electrical stimulation (tES, including tDCS, tACS, and tRNS), and transcranial ultrasound stimulation (tUS), so that tolerability can be recorded consistently and compared across methods.

1.1 What the instrument is for

The NIBS-SES is intended for clinical care and research, including studies in healthy participants.

- **Routine monitoring** of individual patients or participants receiving one or many stimulation sessions.
- **Systematic adverse-effect documentation** in clinical care and in research, using a common vocabulary of side effects.
- **Cross-modality comparison** of tolerability between TMS, tES, and tUS within or across studies.
- **Repeated-measures tracking** of how side effects evolve over a treatment course.
- **Separate recording of within-session and between-session effects**, distinguishing sensations that occur during stimulation from those that appear in the hours or days afterwards.
- **Documentation of factors that can modify the response to stimulation**, such as concomitant medication and sleep, recorded alongside each session so side-effect changes can be read in context.
- **Pragmatic repeated administration**, supported by listing only the most prevalent, evidence-based side effects by name and collecting rarer events through open-text fields, so the form stays short enough to use after every session.

1.2 What it is not

The NIBS-SES is a *screening and monitoring* instrument. It does not replace clinical judgement, a medical examination, formal adverse-event adjudication, or safety-screening/eligibility checklists (for example, contraindication screening prior to TMS). A response on the scale is a prompt for clinical attention where warranted, not a diagnosis.

1.3 How it was developed

The item pool was derived from empirical session data and a structured literature review of adverse effects across TMS, tES, and tUS. The initial pool drew on 851 TMS sessions together with adverse effects reported in the literature, and was then refined through structured expert ratings. Longitudinal analyses supported feasibility for repeated administration. The final instrument separates effects occurring during stimulation from those occurring between sessions. Full development details are described in the primary publication (see Section 9).

2. Structure of the questionnaire

The NIBS-SES organizes side effects along two axes. Understanding both is essential to administering and scoring it correctly.

2.1 Acute vs. delayed effects

The instrument distinguishes:

- **Acute effects** are sensations or effects that occur *during* the stimulation itself (e.g. scalp tingling, muscle twitching sensation, discomfort under the electrode/coil).
- **Delayed effects** occur between sessions, in the hours or days after stimulation (e.g. headache, fatigue, mood changes).

Keeping these separate matters clinically: a sensation felt only during a pulse train is a different phenomenon, with different implications, from a headache that emerges the next morning. Do not merge the two when recording or reporting.

2.2 Two responses per side effect: occurrence AND unpleasantness

For each side effect, the NIBS-SES collects **two distinct pieces of information**. These are not interchangeable:

Response	Question asked	What it records
Occurrence	"Please mark if occurred."	Whether the side effect was present at all (yes / no).
Unpleasantness	"How unpleasant was the sensation?"	Given that it occurred, how burdensome / bothersome it was (a graded rating).

The occurrence mark captures **incidence** (how often a side effect happens). The unpleasantness rating captures **tolerability / burden** (how bad it is when it does happen). Both are needed to describe a side-effect profile properly.

2.3 Critical rule: an occurrence mark is not the same as a zero unpleasantness rating

> **Do not use a "0 / not at all" unpleasantness rating to mean "did not occur."**
>
> The occurrence mark ("Please mark if occurred") and the lowest point of the unpleasantness scale ("not at all unpleasant") mean different things and must be recorded independently.
>
> If a rater leaves the occurrence mark blank and instead writes 0 on the unpleasantness scale, the data become ambiguous: it is then impossible to tell whether **(a)** the side effect never happened, or **(b)** it did happen but simply was not bothersome. These are clinically and statistically different situations, and conflating them corrupts both the incidence and the tolerability estimates.

The rule in practice:

1. **Complete the occurrence response first** for every listed side effect. This is the primary yes/no record.
2. **Rate unpleasantness only for effects that occurred.** A side effect that did not occur has no unpleasantness value, so leave that rating empty (see Section 5.3 on data export coding).
3. **A "0 / not at all unpleasant" rating is valid and meaningful. It says** "this effect occurred but did not bother me." Use it only when the occurrence mark is also set.

Put differently: *occurrence tells you if it happened, and unpleasantness tells you how bad it was. A zero can only ever answer the second question, never the first.*

> **Quick reference**
>
> Occurred, not bothersome → mark occurrence + unpleasantness = 0.
>
> Occurred, bothersome → mark occurrence + unpleasantness = the appropriate rating.
>
> Did not occur → leave occurrence unmarked + leave unpleasantness blank.

3. Before you begin

3.1 Choose the right version

- Format: paper-and-pencil, REDCap, Qualtrics, or SoSci Survey (see Section 5).
- Language: Currently 15 professionally translated and expert-reviewed languages are available (see Section 8).
- Modality: the same form is used for TMS, tES, and tUS. Record which modality and parameters were used alongside the questionnaire.

3.2 When to administer

The NIBS-SES is designed for repeated, session-by-session administration. A typical schedule captures acute effects immediately after each session and delayed effects at the start of the following session (covering the interval since the last one). In accelerated protocols with several sessions per day, the delayed section refers to the interval since the previous session, regardless of its length. Fix your schedule in advance and apply it consistently, because timing affects which effects are captured. Record the exact schedule you adopt:

- Acute items collected: **immediately post-session**
- Delayed items collected: **at start of next session, referencing the inter-session interval**

3.3 Who completes it

The NIBS-SES is a self-report instrument: wherever possible the patient/participant records their own experience, since unpleasantness is inherently subjective. Where a rater assists, they should transcribe the respondent's report without interpretation. Decide and document your mode of administration (self-completed vs. interview-assisted) and keep it consistent across sessions. No baseline information is required beyond the identifiers in the header. The questions on substance use, sleep, and medication changes are context variables that participants answer from their own knowledge.

4. Step-by-step administration

1. **Identify the record**: enter participant/patient ID, session number, date, modality, and stimulation parameters.
2. **Work through the context questions** on substances taken in the last 24 hours, any change in medication in the last 24 hours with substance and dose, and sleep duration and quality of the previous night. Read a reported change against the adverse-effect ratings before and after it. The entry is a self-report and needs no clinical records. Where records are available, check it against them. Decide in advance how such entries enter your analysis.

3. **Explain the two-part response** to the respondent: first they mark whether each effect occurred, then, only for those that occurred, they rate how unpleasant it was.
4. **Work through the delayed section** (effects between sessions), except at first session.
5. **Work through the acute section** (effects during stimulation).
6. **Capture any "other" effects** in the free-text field if present.
7. **Review for completeness**: check there is no item marked as occurred but left without an unpleasantness rating (see Section 2.3).
8. **Act on clinically significant reports** per your safety pathway (Section 7).

5. Digital versions: Qualtrics, SoSci Survey & REDCap

In addition to the paper-and-pencil form, ready-made digital versions are available in all 16 languages, so the questionnaire does not have to be rebuilt. All files are on OSF: https://osf.io/xz3nb.

5.1 Using the Qualtrics template (.qsf)

1. **Download the Qualtrics template file (.qsf) for the respective language from OSF (link above). There is one file per language.**
2. **Import it**: in Qualtrics, go to Projects → Create project → Survey, choose "Use a project file / Import a QSF file," and upload the .qsf file for your language. This recreates the full questionnaire.
3. **Check** that the location questions appear only when the corresponding effect is marked and that the acute and delayed blocks are separated.
4. **Pilot, then publish**: preview end-to-end, then distribute (anonymous link, authenticated link, or per-session link as your design requires).

5.2 Using the SoSci Survey template (.xml)

1. **Download** the SoSci Survey questionnaire file (.xml) from OSF (link above).
    - **Unlike the Qualtrics version, the SoSci Survey XML file contains all language versions in a single file.**
2. **Create a study** in your SoSci Survey account (soscisurvey.de or your institutional instance).
3. **Import the questionnaire**: use SoSci's questionnaire import function to upload the .xml. This restores the questions, the acute/delayed structure, and the filters/conditions that reveal the unpleasantness rating once an effect is marked as occurred.
4. **Verify filters**: in the questionnaire editor, check that the conditional display of unpleasantness items and any page routing carried over correctly.
5. **Pilot, then activate**: run a test interview, review the exported data structure, then activate.

Exact menu labels differ slightly between SoSci Survey versions. If an option is named differently in your version, consult the SoSci Survey documentation for "import questionnaire."

5.3 REDCap (.zip)

1. **Download** the REDCap instrument file (.zip) for your language from OSF (link above). There is one file per language.

    - **Add only one language version to each REDCap project.**

2. **Open or create** your project. Your REDCap project should be in Development status so you can add instruments.

3. **Import the instrument**: go to Project Setup → Online Designer, choose "Upload instrument ZIP file", and select the .zip file for your language. The NIBS-SES is added as a new instrument, and your existing instruments stay as they are. The upload restores the items, the response codes (stored as 1–5, see Section 5.4) and the branching logic that shows the unpleasantness rating only after an effect is marked as occurred.

    - **Do not use the Data Dictionary upload. It replaces all instruments in a project.**

4. **Check variable names and branching logic:** open the instrument in the Online Designer. Make sure no variable names were changed during import, which can happen if your project already has fields with the same names. Then confirm that the occurrence → unpleasantness branching behaves as intended and that the acute and delayed sections are separated.

5. **Enable as survey**: if participants will fill in the questionnaire themselves, turn on "Enable as survey" for the instrument in the Online Designer. Skip this step if staff will enter the data.

6. **Pilot, then go live**: test the instrument from start to finish, export a test record to check the data structure, then move the project to Production and distribute it (public survey link, participant list, or automated invitations, depending on your design).

5.4 Applies to every digital version

- **Preserve the occurrence-vs-unpleasantness distinction**: the two responses must remain separate variables in the export. Never collapse them into a single field.
- **Preserve acute vs. delayed** as distinct sections/variables.
- **Do not alter item wording or scale anchors** when customizing. You may add metadata fields around the instrument, but changes to the items are adaptations (see Section 9) and must be reported as such.
- **Check data export coding**: confirm how "not occurred" (blank unpleasantness) is represented so it is not silently recoded as 0 during analysis.

Data coding: Qualtrics, SoSci Survey and REDCap store the 0–4 ratings as 1–5 (score = stored value − 1) and yes/no answers as 1 = yes, 2 = no.

6. What to report (recommended)

- Occurrence / incidence: the session prevalence of each side effect, i.e. the number of sessions in which each side effect occurred out of all completed sessions. Participant-level counts should additionally be reported.
- Unpleasantness / burden: the distribution of unpleasantness ratings, computed only over the sessions in which the effect occurred.

- Acute vs. delayed: report the two blocks separately.
- Per modality: keep TMS, tES, and tUS distinguishable if you combine data.

7. Safety monitoring and escalation

The NIBS-SES supports safety monitoring but does not define clinical thresholds. Each site could specify in advance how questionnaire responses feed into care and adverse-event reporting.

- **Define action thresholds**: which effects or unpleasantness levels trigger clinician review, parameter adjustment, or pausing treatment.
- **Route serious events** to your standard adverse-event / serious-adverse-event reporting pathway. The questionnaire complements but does not replace it.
- **Watch trends across sessions**: because the instrument is repeated-measures, rising unpleasantness or new delayed effects over a course are visible and should be reviewed.
- **Document actions taken** alongside the responses so the safety record is auditable.

8. Languages and translations

The NIBS-SES has been professionally translated and reviewed by field experts for conceptual accuracy in the following 15 languages:

Arabic, Basque, Catalan, Dutch, French, Galician, German, Hindi, Italian, Japanese, Korean, Mandarin Chinese, Persian, Portuguese, and Spanish.

- **Use the official translated file** for your language rather than translating item text yourself.
- **Record which language version** each respondent used, especially in multi-site studies.
- **Need a language not listed?** Translation is permitted under CC BY 4.0 (Section 9). Follow good cross-cultural adaptation practice and inform the authors, so the version can be listed. We also invite research groups to validate an existing language version in their population and to report wording issues to the authors. Validated versions will be listed in the version history (Section 11).

9. Citation, License, permitted use, and adaptation

Citing the NIBS-SES correctly is both a scientific norm and a requirement of the CC BY 4.0 license.

9.1 Cite the primary publication

When you use the NIBS-SES, cite the peer-reviewed article that describes it. This is the primary reference, and it carries the full author list:

Selaskowski, B., Koelsch, C., Baeken, C., Blumberger, D. M., Brunoni, A. R., Hurlemann, R., Lerma-Usabiaga, G., Noda, Y., Paz-Alonso, P. M., Rivadulla, C., Sun, J., Venkatasubramanian, G., Sack, A. T., Rossi, S., Valero-Cabré, A., & Kiebs, M. (2026). The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): Development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation. Brain Stimulation, 19(6), 103252. https://doi.org/10.1016/j.brs.2026.103252

Example sentence for the methods-section: “We administered the **[LANGUAGE]** version of the NIBS-SES (**[paper citation]**) via **[FORMAT]** (for example paper-and-pencil or SoSci Survey), instrument version **[X.X]** (**[repository citation]**).”

9.2 Also cite the repository

Following standard data-citation practice, also cite the OSF repository you downloaded the files from. The repository record lists the depositing authors, and full authorship appears in the primary publication above.

APA: Selaskowski, B., & Kiebs, M. (2025). The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES). OSF. https://doi.org/10.17605/OSF.IO/XZ3NB

Vancouver: Selaskowski B, Kiebs M. The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES). OSF; 2025. Available from: https://doi.org/10.17605/OSF.IO/XZ3NB

9.3 License, permitted use, and adaptation

The NIBS-SES is open access under the **Creative Commons Attribution 4.0 International (CC BY 4.0)** license. You may use, share, adapt, and translate it for any purpose, including clinical and commercial contexts, provided you give appropriate credit and cite the original source.

- **Attribution is required** wherever the instrument is used or reproduced (see Section 9).
- **Adaptations must be identified as such**: if you change items, wording, scale anchors, or structure, state clearly that your version is adapted from the NIBS-SES and describe the changes, so it is not mistaken for the original.
- **Translations** are permitted under the same terms. Prefer the official expert-reviewed translations where they exist.

Full license terms: https://creativecommons.org/licenses/by/4.0/.

10. Data protection and record-keeping

NIBS-SES responses are health data. The questionnaire header provides fields for a participant or patient identifier, so a pseudonymous code can be entered in place of directly identifying details, with any key kept separately. Storage, hosting, retention, and consent follow your own institutional and regulatory requirements.

11. Version history

Manual version	Date	Instrument version	Changes
1.0	2026-07-20	1.0	Initial release

Manual version	Date	Instrument version	Changes
1.1	2026-10-11	1.0	Update Manual: Intended use stated; wording on monitoring and validation status; accelerated protocols; update on validate language versions
1.2	2026-11-06	1.1	Section 5.3 & 5.4

12. Contact and resources

Official website: https://www.nibs-ses.com/ · FAQ: nibs-ses.com/nibs-ses/faq

Repository (all languages and digital files): https://osf.io/xz3nb

Contacts:

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For correspondence, please use the corresponding author's email address listed in the accompanying open-access publication describing this questionnaire.

This manual is provided to support correct administration of the NIBS-SES. Where it and the official instrument documentation differ, the authors' official documentation on nibs-ses.com and OSF takes precedence.